

Power2Go Battery MES

Functional wireframe | operational shell, primary navigation, and control-room dashboard

POWER2GO

BATTERY MES

QUICK ACCESS

Dashboard

PRODUCTION

New Production

Active Production

Cell Workflow

Module Workflow

Pack / Final QC

INVENTORY

Cells

BMS

BMU

Modules

Batteries

CONTROL

Traceability

Global scan / search: Cell, Module, Battery, BMS, BMU

LINE 01 ACTIVE | Operator profile

Manufacturing Execution & ERP Control

Current order, line health, quality gates, and material availability at a glance.

AVAILABLE CELLS

12,480

1,024 reserved

FIRST PASS YIELD

98.6%

RELEASED PACKS

246

QR passports ready

ACTIVE ORDERS

18

4 due today

MANUFACTURING PIPELINE

01 Ingest

Supplier manifest
Cell inventory
Production order

02 Build

2D battery twin
Scan cells
Scan BMS / BMU

03 Cell QA

OCV / IR
Grading
Damage history

04 Module QA

Matching
Laser welding
Physical QC

05 Release

Pack test
Final QC
QR passport

file:///C:/Users/abdul/Downloads/power2go-battery/docs/MES_WIREFRAME.html

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Quarantine

Reports

Security

Wireframe overview

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Production Planning and 2D Battery Builder

Create an order, reserve material, and assign physical components into a traceable pack structure.

PRODUCTION

New Production

Order queue

Active battery

Workflow gates

Production order / PO-2026-0042

Save draft Create order

PRODUCT TEMPLATE

P2G-LFP-51KWH

51.2V · 100Ah · 8 modules

QUANTITY

10 packs

Required: 1,280 cells · 10 BMS · 10 BMU

MATERIAL CHECK

READY TO RESERVE

Cells 12,480 · BMS 80 · BMU 54

Order created

Cells reserved

2D twin assignment

Cell workflow

Release passport

2D BATTERY TWIN | BAT-000421

Controller bay

BMS ATTACHED

P2G-BMS-000118

BMU ATTACHED

P2G-BMU-000031

Module 01

Cells 01-08: 8 / 8 ASSIGNED

Matching score: 99.2%

Module 02

Cells 09-16: 6 / 8 PENDING

Scan slots 07 and 08

Actions

Scan barcode / QR

Auto-match cells

Continue to cell workflow

Production path: planning → assignment → QA

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file:///C:/Users/abdul/Downloads/power2go-battery/docs/MES_WIREFRAME.html

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Warehouse Inventory: Cells, BMS, BMU, Modules, Batteries

Dense scan-friendly inventory view with controlled CRUD, filters, status, QR, and trace actions.

WAREHOUSE

Cells

BMS

BMU

Modules

Batteries

Warehouse / BMS inventory

Receive batch + export

AVAILABLE80

RESERVED12

IN PROCESS8

QUARANTINE2

BMS

BMU

MODULES

BATTERIES

Search serial, model, protocol, status

Filter: ALL STATUS | Receive batch

Serial number	Model	Protocol	Status	Actions
P2G-BMS-000118	PACE 51.2V 100A	CAN 2.0B	AVAILABLE	QR Edit Delete
P2G-BMS-000119	PACE 51.2V 100A	CAN 2.0B	ASSEMBLED	QR Trace
P2G-BMS-000120	SEPLOS Mason	RS485	QUARANTINED	QR Resolve

Batch modal

Model, manufacturer, protocol, quantity, serial preview, validation, receive, audit event.

Controller inventory is separate for BMS and BMU

Release QR Passport and Traceability

One released battery view exposes the complete physical genealogy and printable labels.

RELEASED BATTERY QR PASSPORT



BATTERY

BAT-000421



BMS

P2G-BMS-000118



BMU

P2G-BMU-000031



MODULE 01

P2G-MOD-000801

Passport actions: download individual QR labels, print passport, close. Labels include parent battery context.

UNIVERSAL TRACEABILITY

Search scanned identifier

BAT-000421

BATTERY FOUND

Genealogy tree

Supplier batch

└ Cell 01 → Module 01 → Battery 000421

└ BMS 000118

└ BMU 000031

└ Final QC → RELEASED

Detail panel

Serial · status · test records · operator · timestamps · audit events

QR labels must be stable, unique, and traceable to the released battery

Quality, Operations, Reporting, and Security

Supporting control surfaces around the core manufacturing path.

CELL / MODULE / PACK QA

OCV / IR	Grading	Damage	Module QC	Final QC
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Tables capture measurement, grade, defect reason, operator, timestamp, machine, and approval.

- PASS
- BYPASS REQUIRES REASON

MACHINE GATEWAY

Station status, heartbeat, protocol, cycles, success rate, power toggle.

- ONLINE
- BUSY

SECURITY

Users · roles · module permissions · action permissions · active/inactive status · audit trail.

- ADMIN
- WAREHOUSE
- QC

QUARANTINE

Open / resolved / all history

Entity	Reason	Stage	Status	Disposition
BMS-000120	CAN frame loss	BMS test	OPEN	Rework / Scrap / Release

REPORTS AND AUDIT

FPY · welding quality · BMS telemetry rate · OCV distribution · Pareto defects · export audit CSV.

Audit priority: server-side permissions, single state machine, transactional writes, and invariant checks before release.

Recommended next design pass: tablet scanner ergonomics and role-specific action visibility